HARSH JAIN

Bengaluru / Lucknow | +91-9091125822 | harshf9@proton.me
github.com/harsh-f9 | linkedin.com/in/harshf9

AI-native M.Sc. AI/ML student and lifelong learner, passionate about computers since age 9. Clear team communicator building forecasting, retrieval, knowledge graph, and engineering drawing systems.

TECHNICAL SKILLS

Languages & backend: Python, SQL, JavaScript, Bash, FastAPI, Pydantic, REST APIs, Celery, Docker
AI, retrieval & data: PyTorch, XGBoost, LSTM, LangGraph, spaCy, GLiNER, BM25, vector search, OpenCV
AI development: SDKs and APIs, OpenAI Responses and Decisions, Claude Code
Databases & tools: PostgreSQL, SQLite, SQLCipher, Neo4j, Qdrant, Git, Linux, ripgrep, ast-grep

EXPERIENCE

Data Analyst Intern | Climate Resilience Observatory, Govt. of U.P. Sep 2025 - Jun 2026
Lucknow, UP | FloodCast live | github.com/harsh-f9/floodcast

- Architected and deployed **FloodCast**, a flood early-warning system covering **378 gauge stations across 69 districts**, with 7-day, multi-tier risk forecasts.
- Built automated ETL pipelines for five climate and geospatial data sources and a hybrid **LSTM + XGBoost** forecasting workflow.
- Integrated an LLM copilot for natural-language queries over reservoir telemetry, historical anomalies, and forecast risk levels.

Account Manager | Sunrise Traders Pune | Nov 2022 - Apr 2024

- Managed financial accounting, ledger reconciliation, and reporting using Excel, Tally, and BUSY.

SELECTED PROJECTS

Airgap Graph | Offline Entity Graph & Engineering Drawing Intelligence 2026 - Present

- Built an offline-first platform that extracts entities and co-occurrence relationships, with an interactive force-directed graph, neighborhood exploration, and source-level character-offset provenance.
- Developed a blueprint vision pipeline using aspect-preserving scaling, overlapping image tiles, adaptive zoom, and sub-patch localization for architectural and engineering diagrams.
- Added visual proof crops, coordinate mapping, and confidence checks for grounded diagram answers; secured local persistence with encrypted storage and optional SQLCipher.

Retrieval Funnel Toolkit | Codebase Indexing & Architecture Analysis 2026
github.com/harsh-f9/retrieval_funnel_toolkit

- Built a six-stage code analysis pipeline using **ast-grep** and **ripgrep** to produce structured dependency graphs from Git repositories.
- Used Pydantic schema validation and PageRank-based ranking to prioritize architectural context; reported benchmark reduced LLM input from **8.7M to 2.1M tokens (75.86%)**.

EDUCATION & RECOGNITION

M.Sc. Computer Applications (AI/ML), IIIT Lucknow | Aug 2025 - Present
B.Sc. (General), North Bengal St. Xavier's College, Siliguri
Delhi Public School, Siliguri

Meta x Scaler Reinforcement Learning Hackathon: Top 100 finalist, Bengaluru.

IIT JAM Mathematics 2025: Qualified, All India Rank 786.